



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 • STANDARD 6.NOS.1

# Fraction Division Problem Solving

Lesson 6-11 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 3 Enrichment



### TODAY'S OBJECTIVES

**Content Objective:** I can solve real-world problems by writing and solving fraction division equations.

**Language Objective:** I can explain my reasoning using the words model, equation, solution, and reasonableness.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Fraction Division Problem Solving

Model

Equation

Solution

Reasonableness

Inverse operations



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

— Use fraction division to find an unknown number of groups or the size of each group.

2

A drawing or diagram that shows a math situation is a

.

3

A math sentence that says two amounts are equal, with an = sign, is an

.

4

The value that makes an equation true is the .

- 5 Checking whether an answer makes sense for the problem is checking its

.

- 6 Operations that undo each other, like multiplication and division, are

.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**How does setting up a fraction division equation help solve this real-world problem?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Fraction Division Problem Solving** — Use fraction division to find an unknown number of groups or the size of each group.

*Resolución de problemas de división de fracciones — Usar la división de fracciones para hallar una cantidad desconocida de grupos o el tamaño de cada grupo.*

- ☐ **Model** — A picture or math way to show a problem so you can solve it.

*Modelo — Un dibujo o forma matemática de mostrar un problema para resolverlo.*

- ☐ **Equation** — A math sentence with an equal sign showing both sides are the same.

*Ecuación — Una oración matemática con un signo igual que muestra que ambos lados son iguales.*

- ☐ **Solution** — A number that makes the equation or inequality true.

*Solución — Un número que hace verdadera la ecuación o desigualdad.*

- ☐ **Reasonableness** — Checking if your answer makes sense.

*Razonabilidad — Revisar si tu respuesta tiene sentido.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The chemist can run \_\_\_\_ tests because  $5/6 \div 1/12 =$  \_\_\_\_.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** The total amount is the dividend; the size of each test is the divisor.

**Evidence:** Students identify  $2/3$  as the total being split (dividend) and  $1/6$  as the group size (divisor), writing  $2/3 \div 1/6$ .

**Reasoning:** This evidence connects the definition of fraction division problem solving to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The chemist can run** \_\_\_\_ **tests because**  $5/6 \div 1/12 =$  \_\_\_\_.
- **My answer is** \_\_\_\_.

*Mi respuesta es* \_\_\_\_.

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is** \_\_\_\_.
- **I know because** \_\_\_\_.

*Mi afirmación es* \_\_\_\_.

*Lo sé porque* \_\_\_\_.

- **So** \_\_\_\_.

*Entonces* \_\_\_\_.

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Fraction Division Problem Solving
2. **Notes 2:** Model
3. **Notes 3:** Equation
4. **Notes 4:** Solution
5. **Notes 5:** Reasonableness
6. **Notes 6:** Inverse operations
7. **Watch for this mistake:** *A common mistake in Fraction Division Problem Solving is reversing which fraction is the dividend and which is the divisor when setting up the equation from a word problem. For example: a pitcher holds  $\frac{3}{4}$  cup of punch, and each serving is  $\frac{1}{8}$  cup — how many servings? The correct setup is  $\text{total} \div \text{group size} = \frac{3}{4} \div \frac{1}{8} = \frac{3}{4} \times \frac{8}{1} = \frac{24}{4} = 6$  servings. A student who reverses the setup calculates  $\frac{1}{8} \div \frac{3}{4} = \frac{1}{8} \times \frac{4}{3} = \frac{4}{24} = \frac{1}{6}$ , an answer smaller than 1 that can't possibly be a serving count. Before you submit, check: does my answer make sense as a NUMBER OF GROUPS? If the setup gives a fraction less than 1 for a 'how many' question, you likely swapped the dividend and divisor.*
8. **Try It 1:** A. 5 — *The total distance ( $\frac{5}{6}$  mile) is the dividend and the checkpoint spacing ( $\frac{1}{6}$  mile) is the divisor:  $\frac{5}{6} \div \frac{1}{6} = \frac{5}{6} \times \frac{6}{1} = \frac{30}{6} = 5$  checkpoints.*
9. **Try It 2:** A. 3 — *The total ( $\frac{9}{10}$  liter) is the dividend and each vial ( $\frac{3}{10}$  liter) is the divisor:  $\frac{9}{10} \div \frac{3}{10} = \frac{9}{10} \times \frac{10}{3} = \frac{90}{30} = 3$  vials, so the code is 3.*